

Drone Security Assessment Report

Security posture summary for live and simulated drone scans

Assessment Report - generated 2026-04-24 16:05 UTC

Analyst: GPtool

Scan type: Simulated — source: Local fake drone / 127.0.0.1

100

/ 100

ATTACK

One or more attacks are trivially feasible.

Scoring method: Machine Learning (ISOT trained)

ML verdict: Attack (100.0% confidence)

Risk Factors Observed

The ML model produced the score above. These are the rule-based factors observed in the same scan.

- +45 open Wi-Fi (Deauth/Evil_Twin/MITM)
- +40 control port exposed (Injection/Replay)
- +25 telemetry port exposed (Spoofing)
- +25 video port exposed (Replay/surveillance)
- +30 MAVLink detected (MITM/Injection)
- +20 combo: weak Wi-Fi + control port
- +20 combo: weak Wi-Fi + MAVLink

Likely Attack Vectors

ISOT attack categories that are feasible given the observed posture, ranked from most to least feasible.

MITM	Injection	Replay	Spoofing	Deauth	Evil_Twin
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Drone Fingerprint

Model	Ryze Tello (simulated)
Wi-Fi	TELLO-FAKE-001 (Open)
Open Ports	8889, 8890, 11111
Protocols	UDP, MAVLink
Firmware Hint	Simulated firmware: demo-v1.0

Vulnerabilities

Wi-Fi network is open

CRITICAL

Enables: Deauth / Evil_Twin / MITM · **Feasibility:** Easy

SSID 'TELLO-FAKE-001' has no encryption. An attacker on the same channel can deauthenticate the controller, clone the AP (Evil Twin), or silently relay traffic (MITM).

Recommendation: Enable WPA2/WPA3 with a strong passphrase. Avoid flying on open networks.

Port 8889 exposed (DJI Command)

CRITICAL

Enables: Injection / Replay · **Feasibility:** Easy

Default Tello/DJI-style control port exposed over UDP. This enables: Injection / Replay.

Recommendation: Restrict access to the control interface, fly in RF-isolated areas, and firewall non-essential ports.

Port 8890 exposed (Telemetry)

HIGH

Enables: Replay / surveillance · **Feasibility:** Medium

Telemetry stream is reachable and may be unencrypted. This enables: Replay / surveillance.

Recommendation: Restrict access to the control interface, fly in RF-isolated areas, and firewall non-essential ports.

Port 11111 exposed (Video Stream)

HIGH

Enables: Replay / surveillance · **Feasibility:** Medium

Video stream is reachable on a common Tello video port. This enables: Replay / surveillance.

Recommendation: Restrict access to the control interface, fly in RF-isolated areas, and firewall non-essential ports.

Unencrypted MAVLink telemetry

HIGH

Enables: MITM / Injection / Spoofing · **Feasibility:** Medium

MAVLink traffic was observed. The protocol has no built-in encryption or authentication, so commands can be forged (Injection) and telemetry can be rewritten in flight (MITM / Spoofing).

Recommendation: Tunnel MAVLink over an authenticated VPN, or use MAVLink 2.0 signing.

Safe Test Results

Test	Result
Ping reachability	Success (loss: 0.0%)

Round-trip latency	12.5 ms (4 samples)
Diagnostic packet	OK — Simulated diagnostic packet accepted. No movement command sent.

AI Analysis

Demo explanation generated for the fake drone scenario.